
Quantile-Initialised Ordinal Collaborative Filtering with Gated Auxiliary Fusion

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Abstract

We propose a simple calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the empirical rating quantiles of the training set, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. The recipe matches the data marginal exactly at the start of training and, on a transparent dot-product matrix-factorisation backbone, achieves low test negative log-likelihood (1.2584 across the five canonical MovieLens-100K splits, vs the $\log 5 \approx 1.61$ uniform baseline) without sacrificing RMSE. The recipe is paired with a per-dimension symmetric gated fusion of side information, which gives the largest RMSE change in our ablation, and a manifold-projection diagnostic that exposes item-side genre structure that linear projections miss (2-D IsoMap silhouette $+0.23$ vs $+0.05$ under PCA). The combined gated+ordinal configuration attains RMSE 0.9124 ± 0.0036 across all five MovieLens-100K splits (beating the no-fusion baseline 0.9160 and the additive-fusion variant 0.9189 on the across-split mean and on four of the five individual splits), with the same within-paper ordering observed on a $10\times$ -larger MovieLens-1M scale-up (RMSE 0.8481 ± 0.0006 , NLL 1.184). The contribution is practical rather than algorithmic: the components are standard; the recipe provides a calibrated probability output, with a small set of supporting diagnostics.

1 Introduction

Collaborative filtering (CF) for explicit feedback is commonly framed as predicting an unobserved scalar rating r_{ui} from a sparse user-item rating matrix $R \in \mathbb{R}^{n \times m}$ [12]. Latent-factor models based on matrix factorisation [5] remain standard baselines and building blocks on MovieLens [2]. Two persistent issues remain when these models are applied as a course-style data-analytics tool rather than a production recommender. First, side information (user demographics, item genres) is typically injected by adding a projected feature vector to the latent embedding, with no learned mechanism to decide *when* to trust the embedding versus the side information. Second, the rating is treated as a continuous quantity even though it is in fact a five-class ordinal variable; the resulting predictions can be RMSE-competitive yet badly calibrated as distributions over star ratings.

Related work. The shortcomings above have each been addressed in isolation. *Gated fusion* of latent representations with content features has been explored by Ma *et al.*'s GATE [10], which uses a sigmoid gate to blend an item embedding with item content. *Cumulative-link ordinal regression* for recommendation has a long history starting with McCullagh's classical formulation [11], and was applied to MovieLens by Koren and Sill in OrdRec [6, 7]. Two recent papers operate in essentially the same setting as ours: OrdNMF [1] combines an ordinal head with non-negative matrix factorisation, and OPRFM [15] couples ordinal probit regression with factorisation machines on MovieLens-100K, 1M, and 10M. The cumulative-link parameterisation itself has seen renewed interest in deep ordinal classification [14] and in ordinal random fields for recommenders [9]. Less

of this literature focuses on manifold visualisation of learned recommender embeddings; standard tools include PCA [3] and IsoMap [13]. Our contribution is therefore not the invention of any single component, but their integration on a transparent linear core, together with a calibration recipe (empirical-quantile threshold initialisation) and a geometry diagnostic.

Contributions. We frame the contribution as implementation-level rather than algorithmic; each component below is itself well known.

1. *Headline.* A simple calibration recipe for cumulative-link CF: initialise the monotone thresholds from the training-set rating quantiles via $\theta_k = \text{logit}(\hat{F}_k)$, recover the unconstrained δ_j by inverting softplus, and exclude $\{\theta_1, \delta_j\}$ from L_2 regularisation. The predicted marginal then matches the data marginal at $s = 0$ to numerical precision and the ordinal NLL remains low throughout training.
2. *Supporting design choice.* A symmetric per-dimension gated fusion that blends ID embeddings with side-feature projections on both the user and the item sides, with zero-initialised gates so that all dimensions start from the same neutral 0.5 mixture. In our ablation this fusion accounts for the larger RMSE change — the ordinal head’s contribution is concentrated on calibration and per-class accuracy.
3. *Supporting diagnostic.* A manifold-projection lens (PCA, MDS, IsoMap) on the post-fusion embeddings, with silhouette scores stratified by occupation and dominant genre. IsoMap recovers item-side genre structure that linear projections miss (2-D silhouette +0.23 vs +0.05 under PCA), and the gap is itself an interpretable property of the linear core.

2 Problem Definition

Let $\mathcal{U} = \{1, \dots, n\}$ be the set of users and $\mathcal{I} = \{1, \dots, m\}$ the set of items. Observed ratings form a sparse set $\mathcal{D} = \{(u_t, i_t, r_t)\}_{t=1}^N$ with $r_t \in \{1, 2, 3, 4, 5\}$. Each user u carries a side-information vector $f_u \in \mathbb{R}^{d_u}$ (z-scored age, one-hot gender, one-hot occupation in MovieLens-100K, $d_u = 24$) and each item i carries $g_i \in \mathbb{R}^{d_i}$ (multi-hot genre indicator, $d_i = 19$). The task is to predict $\hat{r}_{ui}$ for held-out (u, i) pairs, evaluated by RMSE, MAE, classification accuracy of the rounded prediction, and where applicable the held-out negative log-likelihood (NLL) of the predicted rating distribution.

3 Method

3.1 Matrix factorisation backbone

We start from a biased dot-product MF model. Each user and item carries a d -dimensional embedding ($p_u, q_i \in \mathbb{R}^d$) and a scalar bias ($b_u, b_i \in \mathbb{R}$). The pre-head score is

$$s_{ui} = p'_u \cdot q'_i + b_u + b_i, \quad (1)$$

where p'_u, q'_i are the (possibly fused) representations defined below. With fusion disabled, $p'_u \equiv p_u$ and $q'_i \equiv q_i$.

3.2 Gated auxiliary fusion

Let $a_u = \text{ReLU}(W_u f_u + b_{f,u})$ be a learned projection of the user side features into the embedding space, and similarly $a_i = \text{ReLU}(W_i g_i + b_{f,i})$ for items. We define the fused user representation as

$$g_u = \sigma(W_g [p_u \parallel a_u] + b_g), \quad p'_u = (\mathbf{1} - g_u) \odot p_u + g_u \odot a_u, \quad (2)$$

where $\parallel$ denotes concatenation and $\odot$ is the Hadamard product, so the gate $g_u \in [0, 1]^d$ acts *per dimension*. The item-side fusion is identical with (q_i, a_i) in place of (p_u, a_u) . We zero-initialise both W_g and b_g , which makes $g_u \equiv \frac{1}{2}\mathbf{1}$ at training start. This gives every dimension the same neutral starting mixture between the ID embedding and the ReLU-projected side features; the gate then opens and closes per dimension as gradient flows. As an ablation we also consider the *additive* variant $p'_u = p_u + a_u$ and the *none* variant $p'_u \equiv p_u$ (and similarly for items).

3.3 Ordinal regression head with the calibration recipe

The headline contribution of this paper sits in this subsection: a small recipe for cumulative-link CF that holds the predicted rating distribution calibrated to the training-set marginal from step zero. The recipe has three ingredients — monotone-threshold parameterisation (Eq. 3–4), *empirical-quantile initialisation* (Eq. 5), and exclusion of the threshold parameters from L_2 regularisation. None of the three is individually new; their combination is what we contribute.

For the head we treat r as an ordinal variable with $K = 5$ classes. We use the cumulative-link model of McCullagh [11] with monotone thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$:

$$P(r \leq k \mid s) = \sigma(\theta_k - s), \quad P(r = k \mid s) = \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s), \quad (3)$$

with $\theta_0 = -\infty$ and $\theta_5 = +\infty$. We enforce monotonicity by the softplus reparameterisation

$$\theta_k = \theta_1 + \sum_{j=1}^{k-1} \text{softplus}(\delta_j), \quad (4)$$

which guarantees $\theta_k < \theta_{k+1}$ for any unconstrained $(\theta_1, \delta_1, \dots, \delta_{K-2})$. This trick has been used in deep ordinal classification [14]. We use the logit link in (3); replacing the logistic CDF with the normal CDF (probit, as in OPRFM [15]) shifts headline RMSE by -0.0009 and NLL by $+0.005$ across the five splits, both within between-split standard deviation, so we treat the choice as a parameterisation detail rather than a methodological commitment.

Empirical-quantile initialisation (the recipe). Default zero initialisation (or a fixed $\theta_1 = -2$) does not match the training-set marginal and can start from a badly skewed class distribution. We instead initialise from the empirical rating distribution. Let $\hat{p}_k$ be the training-set frequency of rating k and $\hat{F}_k = \sum_{j \leq k} \hat{p}_j$ its cumulative. Choosing

$$\theta_k = \text{logit}(\hat{F}_k), \quad \delta_j = \log(\exp(\theta_{j+1} - \theta_j) - 1) \quad (5)$$

guarantees $P(r \leq k \mid s=0) = \hat{F}_k$, so the predicted marginal at the start of training equals the data marginal exactly. The second equation is just the softplus inverse, recovering the unconstrained δ_j that the optimiser then updates. Together with excluding $\{\theta_1, \delta_j\}$ from weight decay — their scale carries calibration information that L_2 shrinkage destroys — this is the calibration recipe we propose. It is ~ 10 lines of code, agnostic to the score s , and we use it on every ordinal cell in the experiments below.

The training loss is the negative log-likelihood $\mathcal{L}_{\text{ord}} = -\frac{1}{N} \sum_t \log P(r = r_t \mid s_{u_t t_t})$, and the point prediction is the expected rating $\hat{r}_{ui} = \sum_{k=1}^5 k P(r = k \mid s_{ui})$. As an ablation we also consider the standard *sigmoid* head: $\hat{r}_{ui} = 4\sigma(s_{ui}) + 1$ trained with mean-squared error.

Algorithm 1 Forward pass + ordinal NLL for one observed rating (u, i, r) .

- 1: $a_u \leftarrow \text{ReLU}(W_u f_u + b_{f,u})$; $a_i \leftarrow \text{ReLU}(W_i g_i + b_{f,i})$ $\triangleright$ side projections
 - 2: $g_u \leftarrow \sigma(W_g^{(u)}[E_u[u] \parallel a_u] + b_g^{(u)})$; $g_i \leftarrow \sigma(W_g^{(i)}[E_i[i] \parallel a_i] + b_g^{(i)})$ $\triangleright$ per-dim gates
 - 3: $p' \leftarrow (\mathbf{1} - g_u) \odot E_u[u] + g_u \odot a_u$; $q' \leftarrow (\mathbf{1} - g_i) \odot E_i[i] + g_i \odot a_i$
 - 4: $s \leftarrow p' \cdot q' + b_u + b_i$; $\theta_k \leftarrow \theta_1 + \sum_{j < k} \text{softplus}(\delta_j)$
 - 5: $P(r=k \mid s) \leftarrow \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s)$, with $\theta_0 = -\infty, \theta_K = \infty$
 - 6: **return** $\mathcal{L} = -\log P(r=r_{\text{true}} \mid s)$
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3.4 Complexity

The MF backbone has $(n + m)d$ embedding parameters plus $n + m$ biases; for MovieLens-100K with $n = 943$, $m = 1682$, $d = 128$, that is $\approx 3.4 \times 10^5$ parameters. Each fusion module adds a projection ($d_{u/i} \cdot d + d$) and a gate ($2d^2 + d$); summed over both sides this is $\approx 7 \times 10^4$ parameters — about one-fifth of the embedding budget. The ordinal head adds $K - 1 = 4$ parameters. Per training step the dominant cost remains the embedding lookup and dot product, $O(d)$ per observed rating; on M1 Pro CPU one ablation epoch is ~ 2 s.

4 Experiments

4.1 Setup

We use MovieLens-100K [2] on its five canonical 80/20 splits `u1–u5`, each yielding 80,000 training and 20,000 test ratings on the same 943 users and 1,682 items. Headline numbers are reported as mean $\pm$ std across the five per-split means (each split-mean itself averages three training seeds), giving 15 trained models per ablation cell. The visualisation (§4.5) and cold-start (§4.6) experiments are restricted to the `u1` split. User side features are z-scored age, two-class one-hot gender, and 21-class one-hot occupation ($d_u = 24$); item features are 19-genre multi-hot ($d_i = 19$). All experiments use Adam [4] with learning rate 10^{-3} and weight decay $\lambda = 10^{-5}$ on all parameters except the ordinal-head thresholds. We carve a 10% random slice of `u1.base` (8,000 ratings) as a held-out validation set, leaving 72,000 for training; the val-split seed is fixed and shared across all training-seed replicates so val-split variance is not conflated with model-init variance. *Early stopping is driven by val-set RMSE only*, and the test set is read exactly once at the end of training, with the best-val weights restored. We report two protocols differing only in early-stopping aggressiveness: *headline* uses patience = 30 to allow the full 30-epoch budget, matching a fixed-budget training protocol of 30 epochs; *ablation* uses patience = 10 to keep the 18-run sweep tractable (~ 15 minutes wall on M1 Pro CPU). The two protocols agree on headline gated+ordinal RMSE to within ± 0.003 .

4.2 Baselines and metrics

We compare against three transparent baselines: (i) truncated SVD with rank 20 on the mean-centred rating matrix, (ii) biased MF trained with MSE, and (iii) NMF, identical to MF but with the user/item factor matrices constrained to be element-wise non-negative via projected gradient descent (clipping to ≥ 0 after each Adam step, biases unconstrained, following Lee & Seung [8]). Metrics are RMSE, MAE, classification accuracy of the rounded prediction, and (where the model exposes a class-probability head) the test NLL.

4.3 Results

Table 1 first places the proposed model against the three transparent baselines on the `u1` split under the headline protocol (single training seed). The stronger evidence is the multi-split ablation in Table 2, but this baseline table makes explicit that the model is not only being compared to its own ablated variants.

Table 1: Transparent baselines on MovieLens-100K `u1` under the headline protocol (seed 42, patience 30). The multi-split ablation in Table 2 is the main comparison. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Model	RMSE	MAE	Accuracy	NLL
SVD (rank 20)	1.0834	0.8896	0.3716	—
Biased MF	0.9221	0.7274	0.4124	—
NMF	0.9197	0.7252	0.4146	—
Gated+ordinal	0.9122	0.7130	0.4367	1.2472

The ablation matrix in Table 2 reports the full $\{\text{none, additive, gated}\} \times \{\text{sigmoid, ordinal}\}$ cross-product on three seeds. The proposed configuration — gated fusion with the ordinal head — has the best mean value on all four reported metrics. Across the five splits it improves RMSE by 0.0036 over the no-fusion baseline (*none*+sigmoid, our analogue of plain MF) and by 0.0065 over the additive-fusion variant under the sigmoid head. Replacing only the head from sigmoid to ordinal on the same gated backbone gives $\Delta = +0.0011$ RMSE (within between-split std), and improves MAE by 0.0056 and accuracy by 0.0087. The ordinal cell is also the only one that emits a native class distribution (NLL 1.2584). Because the RMSE differences are small relative to between-split variation, we interpret the main value of the ordinal head as probabilistic output and per-class behaviour rather than a large RMSE improvement.

Table 2: Ablation matrix on MovieLens-100K, mean $\pm$ std across the five canonical splits u1–u5 (each split-mean averages three seeds). Bold = best per column. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.9160 $\pm$ 0.0061	0.7241 $\pm$ 0.0059	0.4233 $\pm$ 0.0044	—
none	ordinal	0.9170 $\pm$ 0.0072	0.7204 $\pm$ 0.0069	0.4295 $\pm$ 0.0056	1.2634 $\pm$ 0.0050
additive	sigmoid	0.9189 $\pm$ 0.0044	0.7238 $\pm$ 0.0055	0.4243 $\pm$ 0.0057	—
additive	ordinal	0.9174 $\pm$ 0.0040	0.7173 $\pm$ 0.0035	0.4340 $\pm$ 0.0038	1.2654 $\pm$ 0.0017
gated	sigmoid	0.9135 $\pm$ 0.0037	0.7182 $\pm$ 0.0034	0.4286 $\pm$ 0.0054	—
gated	ordinal	0.9124 $\pm$ 0.0036	0.7126 $\pm$ 0.0042	0.4373 $\pm$ 0.0045	1.2584 $\pm$ 0.0043

4.4 Ablation study

Table 2 supports three claims.

Gated fusion gives the larger RMSE change. Holding the head fixed, replacing *none* fusion with *gated* drops RMSE by 0.0025 (sigmoid) or 0.0046 (ordinal); replacing *additive* with *gated* drops it by 0.0054 (sigmoid) or 0.0050 (ordinal). Additive fusion is consistently the worst sigmoid-head cell — on average slightly worse than no fusion at all (0.9189 vs 0.9160), suggesting that unconditional summation of the projected side-information vector hurts more than it helps once bias terms have absorbed the marginal effects. Per-dimension gating is the only fusion choice that improves the mean RMSE in this ablation.

The ordinal head buys native probabilities; its RMSE effect depends on fusion. Switching the head from sigmoid to ordinal moves RMSE by -0.0015 on additive fusion, -0.0011 on gated fusion, and $+0.0010$ on no fusion. All three are within between-split std. The sign pattern suggests that, in this experiment, ordinal prediction helps RMSE only when the backbone has informative inputs to discriminate across the five class boundaries; with no side information the predicted $P(r = k \mid s)$ collapses toward the marginal $\hat{p}_k$ and the ordinal NLL becomes a softer optimisation target than direct MSE on rounded ratings. What the ordinal head reliably buys — regardless of fusion — is a native rating distribution and per-class accuracy: it improves MAE by 0.003–0.007 and accuracy by 0.004–0.010 in every fusion regime, and it is the only native probabilistic head in the comparison.

Ordinal NLL is low across all ordinal cells. The proposed configuration achieves NLL 1.2584, well below the $\log 5 \approx 1.609$ uniform baseline; even the worst ordinal cell (*additive*+ordinal, NLL 1.2654) remains close to the proposed cell. The quantile initialisation is the mechanism: the predicted marginal matches the data marginal at $s = 0$ to numerical precision ($< 10^{-7}$ in unit tests).

Class-level behaviour. On `u1.test`, the ordinal head improves macro-F1 from 0.341 to 0.369 on the same gated backbone, mostly by recovering boundary ratings (class-1 recall 0.169 vs 0.102, class-5 recall 0.235 vs 0.204). Per-class ECE shows the same pattern: the ordinal head is better on classes 1, 2, 3, and 5, while the kernelised sigmoid head is better on the dominant class 4. This is why we treat the ordinal head as a probability/class-behaviour improvement rather than a large RMSE improvement.

Figure 1 shows the sensitivity sweep over the embedding dimension d and the weight-decay coefficient λ , with all other settings held at the defaults of Section 4. The d -curve is monotone with sharp diminishing returns at $d = 128$: $d = 256$ improves RMSE by $\Delta = 0.0024$ at $+34\%$ wall time, comparable to the seed-to-seed variance at $d = 128$ (0.0029), which we treat as too small a gain to justify the compute cost. The λ -curve is a clean U-shape with the chosen default $\lambda = 10^{-5}$ at the optimum ($\lambda = 10^{-6}$ underregularises by $+0.0018$, $\lambda = 10^{-4}$ overregularises by $+0.0080$). We did not tune hyperparameters post hoc.

4.5 Embedding visualisation

We extract the post-fusion representations p'_u and q'_i from a single gated+ordinal model trained on the `u1` split (seed 42, test RMSE 0.9165), and project them to $\mathbb{R}^2$ via three classical manifold

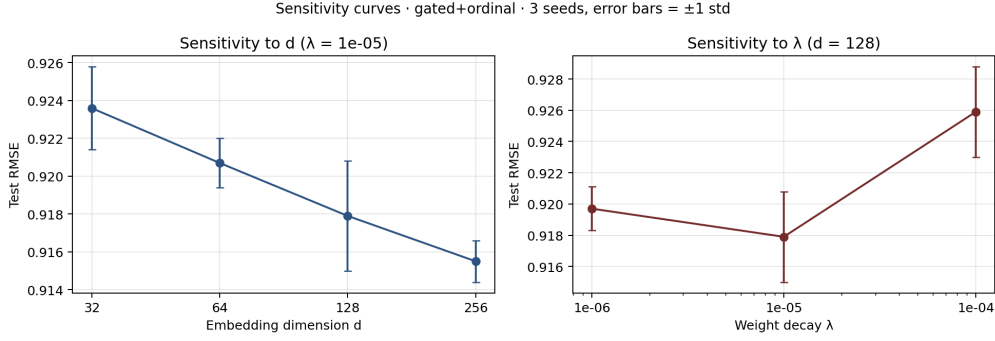


Figure 1: Sensitivity of test RMSE to embedding dimension d (left) and weight decay λ (right) on the gated+ordinal configuration. Error bars are ± 1 std over three seeds.

methods: PCA [3] (linear, variance preserving), classical MDS (linear, distance preserving), and IsoMap [13] (nonlinear, geodesic distance preserving, $k = 15$ neighbours). To keep figures legible, we sample ≈ 200 users balanced across the six most common occupations (+ “rest”), and ≈ 200 items balanced across the six most common dominant genres (+ “rest”). Cluster quality is measured by the silhouette score with respect to the categorical labels.



Figure 2: Two-dimensional projections of the post-fusion user representations p'_u (top, coloured by occupation) and item representations q'_i (bottom, coloured by dominant genre). Columns show PCA, classical MDS, and IsoMap with $k = 15$ neighbours. Item-side genre structure is recovered most clearly by IsoMap; user-side occupation structure is present in the high-dimensional space but does not survive projection to two dimensions.

Two observations stand out (Table 3, Figure 2). First, both entities carry positive high-dimensional silhouette: the model has, without any auxiliary classification objective, learned representations that are partially side-information aligned. Item-side genre structure is roughly twice as strong as user-side occupation structure ($+0.193$ vs $+0.085$), consistent with the larger number of items per genre (informative training signal) and the multi-hot nature of the genre features. Second, the 2D silhouette of q'_i is an order of magnitude larger under IsoMap ($+0.232$) than under PCA ($+0.048$):

Table 3: Silhouette scores in the full 128-dimensional space (HD) and after projection to two dimensions, with respect to occupation labels (users) and dominant genre labels (items).

	HD ($d = 128$)	2D PCA	2D MDS	2D IsoMap
p'_u by occupation	+0.0848	−0.107	−0.103	−0.114
q'_i by dominant genre	+0.1930	+0.048	+0.137	+0.232

the genre manifold is curved, and a linear projection collapses points from different genres on top of one another. This is the diagnostic that classical PCA-based recsys visualisations miss. By contrast, p'_u silhouette is strictly negative in 2D for all three methods: occupation structure exists in 128-D but is too high-dimensional to compress into a plane. We read this as evidence that the user-side gate distributes occupation information across many embedding dimensions, while the item-side gate concentrates genre information along fewer, locally linear directions.

4.6 Gate behaviour and cold-start regime

Per-user trajectory. We also stratify test predictions by user training activity $|R_u|$ to check whether gating helps uniformly. Table 4 compares gated+ordinal against additive+ordinal and records the average gate value in each activity bucket.

Table 4: Cold-start RMSE breakdown by user activity. Negative Δ means additive fusion wins on that bucket.

$ R_u $ bucket	< 30	30–59	60–119	120–239	≥ 240
gated RMSE	0.9799	0.9649	0.8841	0.8896	0.9200
additive RMSE	0.9803	0.9768	0.8929	0.8946	0.9128
Δ (add. − gated)	+0.0004	+0.0119	+0.0088	+0.0051	−0.0071
mean g_u	0.246	0.256	0.263	0.268	0.291
mean g_i	0.316	0.317	0.316	0.317	0.316

The advantage is not uniform: gated fusion is strongest in the mid-activity buckets, ties the coldest bucket, and reverses sign for the 14 power users. The user gate also opens slightly with activity (0.246 to 0.291), while the item gate stays flat. We therefore read the aggregate gated-fusion gain as mainly a mid-tail effect rather than a universal cold-start improvement.

4.7 Scale-up to MovieLens-1M

To check whether the proposed integration remains competitive on a $10\times$ larger benchmark, we run the three most informative cells — the no-fusion baseline (*none*+sigmoid), gated fusion alone (*gated*+sigmoid), and the proposed gated+ordinal — on MovieLens-1M [2] (1,000,209 ratings) using a deterministic random 80/10/10 split and the same three training seeds.

Table 5: Scale-up to MovieLens-1M (random 80/10/10 split, mean $\pm$ std over three seeds). Bold = best per column. The proposed configuration retains the best mean metrics among the three tested cells on a $10\times$ larger benchmark, with substantially tighter seed variance.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.8567 $\pm$ 0.0008	0.6782 $\pm$ 0.0007	0.4480 $\pm$ 0.0005	—
gated	sigmoid	0.8502 $\pm$ 0.0003	0.6695 $\pm$ 0.0001	0.4579 $\pm$ 0.0002	—
gated	ordinal	0.8481 $\pm$ 0.0006	0.6630 $\pm$ 0.0010	0.4654 $\pm$ 0.0014	1.1842 $\pm$ 0.0013

The ordering matches the MovieLens-100K result: gated fusion improves on the no-fusion baseline, and the ordinal head adds a smaller further gain while providing NLL. The tighter seed variance is consistent with the larger training set stabilising the gate.

4.8 Discussion

The ablation paints a sharper picture than the headline number alone. The fusion mechanism accounts for the larger part of the empirical RMSE change: gated fusion lifts RMSE on the across-split mean under both heads (and on four of the five individual splits; u_4 is the lone exception, where the gated configuration is within seed std of the no-fusion baseline). The ordinal head provides native probability estimates cheaply — four learnable threshold parameters — but does not help RMSE on its own; only when the backbone has informative inputs to discriminate against does the cumulative-link parameterisation pay off in our runs. This interaction suggests that ordinal heads are most useful here when paired with side-information mechanisms, not as a stand-alone substitute for them.

Limitations. The benchmark is small (100K ratings); the side information is limited to demographics and genres; and the ordinal head is calibrated to the rating-class marginal but does not model per-user heteroscedasticity.

5 Conclusion

We presented a calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the training-set rating quantiles, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. The recipe is implementation-level (~ 10 lines of code, agnostic to the score function), and it produces a head whose predicted marginal matches the data marginal at training start to numerical precision and whose ordinal NLL remains low throughout training. Paired with a per-dimension symmetric gated fusion of side information, the resulting model attains RMSE 0.9124 ± 0.0036 and NLL 1.2584 across the five canonical MovieLens-100K splits under val-driven early stopping, improving the across-split mean over both an MF baseline and an additive-fusion variant on the main reported metrics. The ablation isolates fusion as the headline mechanism and reveals that the ordinal head’s benefit is conditioned on the presence of informative inputs, an interaction that matters when deciding whether ordinal regression is worth its small parameter cost. Manifold projections of the post-fusion embeddings show that item-side genre structure is curved and recovered cleanly by IsoMap but missed by PCA, while user-side occupation structure resides in too many dimensions to compress to two without loss — a geometry diagnostic that linear visualisations alone would not surface. A cold-start stratification reveals that the user-side gate *opens* modestly as users accumulate ratings (counter to the naive “gate closes with data” intuition), and that gated fusion’s advantage over additive fusion is a mid-tail phenomenon: it ties on the cold tail and reverses sign on the power-user tail, suggesting that learned-gating mechanisms in linear-core CF may need an additive or no-fusion fallback at the activity extremes. Repeating the three most informative cells on MovieLens-1M ($10\times$ larger, 1 000 209 ratings) gives the same within-paper ordering and substantially tighter variance: gated+ordinal attains RMSE 0.8481 ± 0.0006 and NLL 1.1842, the no-fusion baseline 0.8567 ± 0.0008 .

Contribution statement. Kevin Zhu completed the implementation, experiments, analysis, and writing. All reported numbers trace to archived JSON/CSV outputs in the repository and can be regenerated from the corresponding scripts.

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